

OpenH-4D: Collaborative Collection of 4D Cardiac and Respiratory Data

1. Introduction

The OpenH-4D initiative is a new dataset collection effort to accelerate the development of foundational models of dynamic physiology.

Building on the success of the OpenH-Embodiment video collection initiative, **our goal is to align leading institutions to collect 4D data from at least 20,000 subjects, where that data is comprised of patient details as well as sequences of 3D images capturing cardiac and/or respiratory motion.**

The collected datasets will be made freely available for academic and commercial applications and used to train and validate physics-informed and generative AI models of physiological motion. This data will also be well suited for investigating a diversity of applications, including: detecting imaging biomarkers that inform parameters for patient-specific physics, training diagnostic models that incorporate physiological data, and simulating multiple clinical signals such as EKGs and spirometry as well as simulating motion artifacts in 3D images.

NVIDIA will sponsor OpenH-4D by pre-processing all submissions as well as enabling free and open access to the OpenH-4D dataset, foundation model weights, and training code upon project completion.

2. Steering Group

2.1. Leadership

Academic Lead	Alison Marsden, PhD	Stanford
Generative AI Lead	Can Zhao, PhD	NVIDIA
Industry Lead	Mark Palmer, MD, PhD	Ansys, Part of Synopsys
Co-Chair	Mathias Uebernath, PhD	John's Hopkin's University
Clinical Lead	Matthew Jolley, MD	Children's Hospital of Philadelphia
Academic Lead	Paul Segars, PhD	Duke University
Co-Chair	Stephen Aylward, PhD	NVIDIA
Academic Lead	XXXX	XXXX (GenAI expert)

2.2. Founding Members

Ryan Moore, MD
Jeff Bodner, PhD

Cincinnati's Children's Hospital Medical Center
Medtronic

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3. License

The OpenH-4D dataset and model weights will be publicly available on Hugging Face under the CC BY 4.0 license.

4. What is a foundation model

A **foundation model for physiological motion** is a large AI model pre-trained on diverse 4D data to learn the general principles of how anatomy moves, deforms, and responds to physiological conditions. Instead of being built for one motion-estimation task or one imaging modality, it provides a learned representation of anatomy, dynamics, mechanics, and context that can be adapted to specific patient conditions or to many downstream clinical, simulation, and digital-twin tasks with relatively little task-specific data.

5. Why does data matter for Foundation Models?

Data is central to the success of foundation models because their goal is to learn broad, transferable representations rather than solutions tailored to a specific subpopulation. This requires training on **large and diverse datasets** that extend beyond a single modality, imaging protocol, or patient demographic so the model can capture general structure in the data. With sufficient scale and diversity, the model can learn underlying relationships in the data, allowing it to approximate aspects of the **physical or generative processes** that produce the observations. This enables the model to generalize more effectively across tasks, settings, and downstream applications.

6 . Objectives & Outcomes

1. **Data Alignment** – Curate a minimum of **20,000 sequences of high-quality, 4D, CT, MRI, or Ultrasound scans** of cardiac or lung respiratory motion.
2. **Generalist Foundational Models** – Enable training of large, AI models (multi-modal, multi-task) that can be fine-tuned to subpopulations with $\leq 10\%$ of current data requirements.
3. **Open-Source Release** – Publish cleaned and pre-processed datasets and model weights under a permissive license (CC-BY) no later than March, 2027.

4. **Community Enablement** - Enable the physiological simulation community, both research and commercial, to create dynamic cardiac and respiratory applications based on foundation model weights.

7. Scope of Work

Participating teams may submit 4D CT, MRI, and Ultrasound imaging data for **one or more** of the following task groups:

- Cardiac motion
- Lung respiratory motion

In the future, we anticipate expanding the scope to other modalities (e.g., dynamic x-ray fluoroscopy) and other anatomies and physiological functions (e.g., brain perfusion).

8. Proposal and Data Requirements

Each proposal must describe the data to be contributed, with the following details:

- Executive Summary
 - High-level description of the data and its source (500 words or less)
 - Table summarizing key quantifications of the data
 - Organ (Heart/Lung)
 - Typical clinical justification for why image was acquired (e.g., lung cancer treatment planning)
 - Relevant 4D imaging protocol (e.g., MRI cine)
 - Number of patients
 - Typical number of time points in each 4D sequence
 - Typical imaging resolution
- Data Details
 - Imaging
 - Format: DICOM or other
 - Number of time points
 - Coverage of organ of interest: Full or partial
 - Contrast: Yes / No
 - Annotations: Segmentations, landmarks, categorizations
 - Modality specifics
 - CT
 - Low Dose: Yes / No
 - Resolution
 - In-plane: 1 mm or less
 - Slice spacing:
 - Cardiac: 2mm or less

- Lung: 3mm or less
- MRI
 - Cine Resolution
 - Other sequences included
- Ultrasound
 - Source: 4D TEE, 4D Percutaneous, other
- Patient information
 - Demographics: Age, gender, weight, height, ...
 - Disease state: AFib, PE, cancer, ...
 - Vital signs at acquisition: BP, respiratory rate, pulse
 - Relevant treatment history: Post surgery, post radiation therapy, ...
 - Reason for scan: Screening, Surgical planning, ...
- Approvals
 - Privacy safeguards
 - IRB and/or other institutional documentation approving acquisition and release
 - Confirmation of approval for public release under the CC-BY license.
- Team Qualifications
 - Project lead, contact information, and 4-page-or-less CV
 - Technical support lead, contact information, and 4-page-or-less CV
 - List of relevant team members and their roles
 - Public, free-access links to at most 5 related publications from the team
- Author List and Citation
 - Authors to be listed on the planned publications (see Section 9, Contribution Benefits)
 - Paper to be cited relevant to the data contributed.
- Schedule
 - Delivery technology and schedule, in consideration of the timeline in Section 7

Pre-proposals may be submitted to receive early feedback on the relevance of the data to the OpenH-4D effort. These should include the Data Details outlined above.

Full proposals should be submitted in PDF format. Proposals should not exceed 5 pages, excluding CVs.

Proposal submission (pre- and full proposals) should be submitted using this form **XXXXXXXXXX**

Review criteria include (a) completeness of application, (b) uniqueness of the data, (c) size and consistency of the data, and (d) suitability of the data for the proposed AI foundation models. We will also attempt to achieve a 40/40/20 split of the data for CT, MRI, and ultrasound and a 50/50 split of lung and cardiac data.

7. Timeline

Milestone

Date

RFP released	September 27, 2026
Public Proposal submission deadline	November 1, 2026
Data collection window	October 1 – December 31, 2027
Data Cleanup/Standardization	January 1 – February 1, 2027
Model training & validation	February 1 – March 15, 2027
First public release of full dataset and models	April, 2027

8. Eligibility & Partnership Models

- Academic institutions, start-ups, and industrial healthcare firms are welcome.
- International participation is encouraged.
- Consortia of multiple organizations may submit a joint proposal with a single lead.

9. Contribution Benefits

Contributing teams will be named co-authors on both the OpenH-4D dataset publication and the follow-up publication describing the resulting OpenH-4D foundation models.

Contributing teams will also receive early-access evaluation checkpoints of the OpenH-4D foundation model, and the dataset itself, for their own downstream experiments.

Contributing teams will also receive recognition during the joint dataset and model releases at NVIDIA's GTC 2027 conference.

10. Data Governance, Ethics & Compliance

- **Patient Privacy:** All human-subject data must be de-identified to HIPAA Safe Harbor standards or equivalent.
- **Regulatory:** Proposers are responsible for obtaining IRB/ethics approval where required.
- **Licensing:** Final datasets will be released under CC BY 4.0; all assets must be free of third-party IP encumbrances.

11. Contact Information

- Technical Questions
 - OpenH-4D organizers: openh.data+4d@gmail.com
- Administrative Questions
 - Stephen Aylward: saylward@nvidia.com